A Function Field Euclid–Mullin Sequence: Unconditional Sieve Results and Orbit Barriers

With Formal Verification in Lean 4

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Abstract

We study the function field analog of the Euclid–Mullin sequence over $\mathbb{F}_p[t]$, where monic irreducible polynomials play the role of primes. Our main result is an unconditional almost-all theorem: for density-one monic squarefree starting polynomials, the greedy factorization walk reaches every monic irreducible target. The proof proceeds via a six-step analytic number theory chain—character sum cancellation, a prime polynomial theorem in arithmetic progressions, forbidden class divergence, sieve product vanishing, confined density decay, and pigeonhole capture—each step of which is *unconditional* over $\mathbb{F}_p[t]$, in contrast to the integer setting where the chain requires the Weighted Prime Number Theorem in arithmetic progressions.

We prove that the orbit-specificity barrier—the gap between population-level equidistribution and orbit-level equidistribution—is the sole remaining obstacle to the full function field Mullin Conjecture, and that this barrier is structurally identical to the integer case. Three formally verified dead ends confirm that the Weil bound, large monodromy, and subgroup escape are each insufficient to close this gap.

We also prove an unconditional exclusion criterion: for primes $p \equiv 2 \pmod 3$ with $p \geq 5$, the autonomous map $f(w) = w(w+1)$ on $\mathbb{F}_p$ has no preimage of -1, giving a structural obstruction to degree-one capture in the function field setting with no integer analog.

The entire development comprises 17 files and 7,372 lines of Lean 4 code, with zero `sorry` axioms, verified against Mathlib.

Contents

1 Introduction **3**
- 1.1 The Euclid–Mullin sequence . . . 3
- 1.2 The function field analog . . . 3
- 1.3 Why study the function field case? . . . 3
- 1.4 Main results . . . 4
- 1.5 Orbit barrier thesis . . . 4
- 1.6 Machine verification . . . 4
- 1.7 Organization . . . 5

2 Background **5**
- 2.1 The polynomial ring $\mathbb{F}_p[t]$ . . . 5
- 2.2 Counting monic polynomials . . . 5
- 2.3 The necklace identity . . . 5
- 2.4 The FF-EM sequence: basic properties . . . 6

2.5 The autonomous map perspective . . . 7

3 The Unconditional Sieve Chain **7**
3.1 Overview: the integer chain versus the function field chain . . . 7
3.2 Step 1: Character sum cancellation . . . 8
3.3 Step 2: PNT in arithmetic progressions . . . 8
3.4 Step 3: Forbidden class divergence . . . 8
3.5 Step 4: Sieve product vanishing . . . 9
3.6 Step 5: Population sieve confinement decay . . . 9
3.7 Step 6: Almost-all generalized mixed MC . . . 10
3.8 The full chain: a single landscape theorem . . . 10
3.9 The factor tree and structural prerequisites . . . 10
3.10 The stochastic MC and the phase transition . . . 11
3.11 The bootstrap: from dynamical hitting to FF-MC . . . 11

4 The Φ_3 Exclusion Criterion **12**
4.1 The autonomous map and degree-one targets . . . 12
4.2 Fixed points and absorption . . . 12
4.3 The preimage of -1 . . . 12
4.4 Absence of cube roots of unity . . . 12
4.5 The main exclusion theorem . . . 13
4.6 Interpretation: a structural obstruction . . . 13
4.7 Comparison with the integer case . . . 14
4.8 The cyclotomic structure: Dead End #129 . . . 14

5 The Orbit-Specificity Boundary **14**
5.1 Population versus orbit equidistribution . . . 14
5.2 Dead End #127: Weil bound insufficient . . . 15
5.3 Dead End #129: Large monodromy structurally false . . . 15
5.4 Dead End #130: Generation does not imply coverage . . . 15
5.5 The orbit barrier thesis . . . 16
5.6 Setting independence . . . 16
5.7 Summary comparison with the integer case . . . 17

6 Formalization Notes **17**
6.1 Overview . . . 17
6.2 File inventory . . . 17
6.3 The necklace identity proof . . . 18
6.4 Key Mathlib dependencies . . . 18
6.5 Open propositions . . . 19
6.6 Verification and reproduction . . . 19

7 Open Problems **19**

1 Introduction

1.1 The Euclid–Mullin sequence

The *Euclid–Mullin sequence* is defined by $a_0 = 2$ and

$$a_{n+1} = \text{least prime factor of } P(n) + 1, \qquad P(n) = a_0 \cdot a_1 \cdots a_n.$$

The first several terms are $2, 3, 7, 43, 13, 53, 5, 6221671, \ldots$ (OEIS A000945). The construction mimics Euclid's proof of the infinitude of primes: each $P(n) + 1$ is coprime to all previously seen primes, so a_{n+1} is always a *new* prime. Mullin [Mul63] asked whether every prime eventually appears in this sequence.

Open Problem 1 (Mullin's Conjecture, 1963)**.** For every prime q, there exists n such that $a_n = q$.

Despite substantial computational evidence—Guy and Nowakowski [GN75] computed 51 terms, and subsequent work has extended this further—Mullin's Conjecture remains open. The prime 41 has not appeared in the first several hundred computed terms, and Cox and van der Poorten [CvdP68] raised the possibility that it might never appear.

1.2 The function field analog

Over the polynomial ring $\mathbb{F}_p[t]$ (with p a prime number), monic irreducible polynomials play the role of primes. We define the *function field Euclid–Mullin sequence* (FF-EM) as follows.

Definition 1.1 (FF-EM Sequence)**.** [`Lean`: `FFEMData`] Fix a prime p. The FF-EM data consists of:

- $\mathrm{ffSeq}(0) = t$ (the "smallest" monic irreducible),
- $\mathrm{ffProd}(0) = t$ (the running product),
- $\mathrm{ffSeq}(n + 1)$ is a monic irreducible factor of $\mathrm{ffProd}(n) + 1$ of minimal degree,
- $\mathrm{ffProd}(n + 1) = \mathrm{ffProd}(n) \cdot \mathrm{ffSeq}(n + 1)$.

Open Problem 2 (Function Field Mullin Conjecture)**.** [`Lean`: `FFMullinConjecture`] For every monic irreducible polynomial $Q \in \mathbb{F}_p[t]$, there exists n such that $\mathrm{ffSeq}(n) = Q$.

Two structural differences from the integer setting are immediate. First, among irreducible polynomials of the same degree, there is no canonical total ordering; the minimality condition in Definition 1.1 is on *degree* only, so the successor $\mathrm{ffSeq}(n + 1)$ may be chosen among several competitors. Second, the unit group $(\mathbb{F}_p[t]/(Q))^\times \cong \mathbb{F}_{p^{\deg Q}}^\times$ is cyclic of order $p^{\deg Q} - 1$, providing a richer group-theoretic structure than the integer analog $(\mathbb{Z}/q\mathbb{Z})^\times$.

1.3 Why study the function field case?

The function field setting enjoys a decisive analytic advantage over $\mathbb{Z}$: the Weil bound for character sums over $\mathbb{F}_p$-curves is a *theorem* (Weil, 1948 [Wei48]), not a conjecture. This makes population equidistribution of monic irreducibles across residue classes unconditional, in contrast to the integer case where the analogous result requires the Weighted Prime Number Theorem in arithmetic progressions (a consequence of the Generalized Riemann Hypothesis for quantitative error terms, or of Siegel–Walfisz for qualitative versions).

By working over $\mathbb{F}_p[t]$, we can isolate the *orbit-specificity barrier*—the gap between knowing that irreducibles are equidistributed across residue classes (population statistics)

and knowing that the *specific* walk ffProd(n) mod Q visits the residue class -1 (orbit dynamics). Our results demonstrate that this barrier is the *sole remaining obstacle* to FF-MC, and that it is structurally identical to the integer case.

1.4 Main results

We prove four main theorems, all unconditional (zero open hypotheses in the formal verification).

Theorem 1.2 (Almost-All Generalized Mixed MC — Theorem D)**.** *[Lean: `ff_almost_all_unconditional`] Over $\mathbb{F}_p[t]$, the full analytic number theory chain*

$$\textit{Char. Cancel.} \Rightarrow \textit{PNT-in-APs} \Rightarrow \text{FCD} \Rightarrow \text{SPV} \Rightarrow \text{PSCD} \Rightarrow \textit{Almost-All GenMixedMC}$$

is unconditional. For density-one monic squarefree starting polynomials, every monic irreducible is reached by some branch of the factor tree.

Theorem 1.3 (Factor Tree Structure — Theorem A)**.** *[Lean: `theorem_A_factor_tree`] For any valid starting polynomial and any valid mixed selection: the walk explores infinitely many distinct monic irreducibles; the factor pool degree grows to infinity; the selection is injective (no irreducible appears twice); the accumulated product is injective; and the coprimality cascade holds.*

Theorem 1.4 (Stochastic FF-MC — Theorems B and C)**.** *[Lean: `theorem_B_stochastic_mc`] With any positive probability of non-greedy factor selection at each step, every monic irreducible is eventually captured. The phase transition is sharp at $\varepsilon = 0$: deterministic FF-MC is the sole unresolved boundary.*

Theorem 1.5 (Φ_3 Exclusion Criterion)**.** *[Lean: `autonomous_map_landscape`] For $p \equiv 2 \pmod 3$ with $p \geq 5$, the autonomous map $f(w) = w(w+1)$ on $\mathbb{F}_p$ has no preimage of -1. The orbit from any starting value $a \neq -1$ never hits -1.*

1.5 Orbit barrier thesis

Three formally verified dead ends establish the structural boundary of current techniques:

(i) **Dead End #127** (Weil insufficient): The Weil bound gives population equidistribution but cannot constrain orbit dynamics. [Lean: `dead_end_127_weil_not_sufficient`]

(ii) **Dead End #129** (FFLM false): The large monodromy hypothesis is structurally false—the greedy FF-EM construction produces cyclotomic products with abelian Galois groups, reducing monodromy arguments to the Weil bound (circular). [Lean: `dead_end_129_monodromy_dead`]

(iii) **Dead End #130** (Generation $\neq$ coverage): The steps $\{1, 3\}$ generate $\mathbb{Z}/4\mathbb{Z}$ but the alternating walk visits only $\{0, 1\}$, missing $\{2, 3\}$. [Lean: `dead_end_130_generation_ne_coverage`]

Together, these identify the orbit-specificity barrier as not solvable by population statistics (Weil, PE, ANT), not solvable by Galois theory (large monodromy is false), and not solvable by algebraic generation (subgroup escape $\neq$ dynamical hitting). The sole known dissolution is stochastic perturbation ($\varepsilon > 0$). The deterministic case ($\varepsilon = 0$) is the open conjecture.

1.6 Machine verification

The entire development is formalized in Lean 4 using the Mathlib library and compiles with zero `sorry` axioms. The function field module comprises 17 files totaling 7,372 lines of code, with 364 declarations. We discuss the formalization in Section 6.

1.7 Organization

Section 2 introduces the algebraic background: $\mathbb{F}_p[t]$ as a polynomial ring, monic irreducibles, the necklace identity, and the autonomous map perspective. Section 3 presents the unconditional sieve chain culminating in Theorem 1.2. Section 4 develops the Φ_3 exclusion criterion (Theorem 1.5). Section 5 analyzes the orbit-specificity barrier and the three dead ends. Section 6 discusses the Lean formalization. Section 7 collects open problems.

2 Background

2.1 The polynomial ring $\mathbb{F}_p[t]$

Let p be a prime number. The ring $\mathbb{F}_p[t] = \mathbb{F}_p[t]$ of polynomials over the finite field $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$ is a Euclidean domain, hence a principal ideal domain and a unique factorization domain. Monic irreducible polynomials play the role of prime numbers; we shall call them simply *primes* when the context is clear.

We write $\pi_p(d)$ for the number of monic irreducible polynomials of degree d over $\mathbb{F}_p$. The first values for $p = 2$ are $\pi_p(1) = 2$ (t and $t + 1$), $\pi_p(2) = 1$ ($t^2 + t + 1$), $\pi_p(3) = 2$, $\pi_p(4) = 3$.

Proposition 2.1 (Existence of irreducibles)**.** *[Lean: `ffIrredCount_pos`] For every $d \geq 1$ and every prime p, there exists at least one monic irreducible polynomial of degree d over $\mathbb{F}_p$:*

$$\pi_p(d) \geq 1 \quad \textit{for all } d \geq 1.$$

The proof in our formalization uses the Galois field construction: for each $d \geq 1$, the field $\mathbb{F}_{p^d}$ is an extension of $\mathbb{F}_p$ of degree d, and the minimal polynomial of a primitive element is a monic irreducible of degree d.

2.2 Counting monic polynomials

Proposition 2.2 (Monic polynomial count)**.** *[Lean: `card_monic_of_degree`] There are exactly p^n monic polynomials of degree n over $\mathbb{F}_p$.*

Proof. A monic polynomial of degree n has the form $t^n + a_{n-1}t^{n-1} + \cdots + a_0$, and each coefficient a_i can be chosen independently from $\mathbb{F}_p$. In the formal proof, we compose the equivalence between monic degree-n polynomials and the submodule $\mathrm{degreeLT}(n)$ (via `monicEquivDegreeLT`) with the linear equivalence to $\mathrm{Fin}(n) \to \mathbb{F}_p$ (via `degreeLTEquiv`), then compute $|\mathrm{Fin}(n) \to \mathbb{F}_p| = |\mathbb{F}_p|^n = p^n$. □

2.3 The necklace identity

The central combinatorial identity underlying our sieve chain relates the count of monic irreducibles to the total count of monic polynomials.

Theorem 2.3 (Necklace Identity)**.** *[Lean: `necklace_identity_proved`] For every $n \geq 1$,*

$$\sum_{d|n} d \cdot \pi_p(d) = p^n. \tag{1}$$

The name "necklace identity" comes from the combinatorial interpretation: $\pi_p(d)$ counts the number of aperiodic binary necklaces of length d (for $p = 2$), and the identity (1) is

equivalent to the Gauss–Waring formula counting $\mathbb{F}_p$-rational points on the affine line. The identity is classical, attributed to Moreau (1872) and implicit in Gauss's work on cyclotomy.

Proof sketch. The identity (1) is equivalent to the factorization

$$t^{p^n} - t = \prod_{\substack{Q \text{ monic irred} \\ \deg Q \mid n}} Q \tag{2}$$

over $\mathbb{F}_p[t]$. Both sides are monic of degree p^n, so it suffices to show mutual divisibility.

Forward direction. Each monic irreducible Q of degree $d \mid n$ divides $t^{p^d} - t$ (since Q is a minimal polynomial of some element of $\mathbb{F}_{p^d}$), and $t^{p^d} - t$ divides $t^{p^n} - t$ (since $\mathbb{F}_{p^d} \subseteq \mathbb{F}_{p^n}$). The product of distinct monic irreducibles is coprime (by unique factorization), so their product divides $t^{p^n} - t$.

Reverse direction. Let $\alpha \in \mathbb{F}_{p^n}$. Its minimal polynomial Q_α over $\mathbb{F}_p$ is monic, irreducible, and has degree d_α dividing n (by the tower law: $d_\alpha = [\mathbb{F}_p(\alpha) : \mathbb{F}_p]$ divides $[\mathbb{F}_{p^n} : \mathbb{F}_p] = n$). Hence Q_α divides the product, so α is a root of the product. Since all p^n elements of $\mathbb{F}_{p^n}$ are roots of the product, and the product has degree $\sum_{d \mid n} d \cdot \pi_p(d)$, we conclude

$$p^n \leq \sum_{d \mid n} d \cdot \pi_p(d).$$

Combined with the forward bound ($\leq p^n$ from divisibility and degree comparison), equality follows. □

The formal proof in Lean runs approximately 380 lines and makes essential use of Mathlib's Galois theory infrastructure: `GaloisField`, `IntermediateField.finrank_dvd_of_le_right`, `minpoly.irreducible`, and `FiniteField.X_pow_card_pow_sub_X_ne_zero`.

Corollary 2.4 (Upper and lower bounds on $\pi_p(n)$)**.** *[Lean: `necklace_count_mul_le`] For every $n \geq 1$:*

(i) $n \cdot \pi_p(n) \leq p^n$ (the $d = n$ term in (1) is at most the total sum),

(ii) $\pi_p(n) \leq p^n/n$.

A more refined analysis using Mobius inversion gives $\pi_p(n) = \frac{1}{n} \sum_{d \mid n} \mu(n/d) p^d$, yielding the asymptotic $\pi_p(n) \sim p^n/n$ with error $O(p^{n/2}/n)$. For our sieve chain, the crude bound $n \cdot \pi_p(n) \leq p^n$ suffices.

2.4 The FF-EM sequence: basic properties

Proposition 2.5 (Coprimality cascade)**.** *[Lean: `coprimality_cascade`] Let $Q \in \mathbb{F}_p[t]$ be irreducible. Then Q cannot divide both* $\mathrm{ffProd}(n)$ *and* $\mathrm{ffProd}(n) + 1$.

Proof. If $Q \mid \mathrm{ffProd}(n)$ and $Q \mid \mathrm{ffProd}(n) + 1$, then $Q \mid ((\mathrm{ffProd}(n) + 1) - \mathrm{ffProd}(n)) = 1$, contradicting the irreducibility of Q (which has degree ≥ 1). □

Proposition 2.6 (Injectivity)**.** *[Lean: `ffSeq_injective_proved`] The map $n \mapsto \mathrm{ffSeq}(n)$ is injective: no monic irreducible polynomial appears twice in the FF-EM sequence.*

Proof. Suppose $\mathrm{ffSeq}(j) = \mathrm{ffSeq}(k)$ with $j < k$. Then $\mathrm{ffSeq}(j) \mid \mathrm{ffProd}(k-1)$ (since $\mathrm{ffSeq}(j)$ is a factor of the product at index $j \leq k-1$), and $\mathrm{ffSeq}(k) \mid \mathrm{ffProd}(k-1)+1$ (by definition of the recurrence). Since $\mathrm{ffSeq}(j) = \mathrm{ffSeq}(k)$, this polynomial divides both $\mathrm{ffProd}(k-1)$ and $\mathrm{ffProd}(k-1)+1$, contradicting Proposition 2.5. □

Proposition 2.7 (Degree growth)**.** *[Lean: `ffProd_natDegree_strict_mono`] The degree of* $\mathrm{ffProd}(n)$ *is strictly monotone:* $\deg(\mathrm{ffProd}(n+1)) > \deg(\mathrm{ffProd}(n))$.

Proof. We have $\mathrm{ffProd}(n+1) = \mathrm{ffProd}(n) \cdot \mathrm{ffSeq}(n+1)$. Since $\mathrm{ffProd}(n)$ and $\mathrm{ffSeq}(n+1)$ are both monic with $\deg(\mathrm{ffSeq}(n+1)) \geq 1$ (as an irreducible polynomial), the degree of the product strictly exceeds that of $\mathrm{ffProd}(n)$. □

2.5 The autonomous map perspective

When all irreducible factors of $\mathrm{ffProd}(n)+1$ have degree 1 (i.e., when $\mathrm{ffProd}(n)+1$ splits completely into linear factors), the walk position $\mathrm{ffProd}(n) \bmod Q$ for a degree-1 target $Q = (t-a)$ follows the autonomous map

$$f(w) = w \cdot (w+1)$$

on $\mathbb{F}_p$. This is because under the assumption of perpetual irreducibility, the residue of $\mathrm{ffProd}(n+1) = \mathrm{ffProd}(n) \cdot \mathrm{ffSeq}(n+1)$ modulo Q depends only on the residue of $\mathrm{ffProd}(n)$ modulo Q.

Definition 2.8 (Autonomous map)**.** [Lean: `ffAutonomousMap`] For a prime p, define $f\colon \mathbb{F}_p \to \mathbb{F}_p$ by $f(w) = w(w+1)$. The *orbit* from a starting value $a \in \mathbb{F}_p$ is the sequence $a, f(a), f^2(a), \ldots$

The map f has the following elementary properties:

- $f(-1) = (-1)(0) = 0$ and $f(0) = 0$: the value -1 maps to the absorbing fixed point 0. [Lean: `ffAutonomousMap_neg_one`]
- $f(w) = -1$ if and only if $w^2 + w + 1 = 0$ (since $f(w) = w^2 + w$): the preimage of -1 is the root set of the third cyclotomic polynomial $\Phi_3(w) = w^2 + w + 1$. [Lean: `ffAutonomousMap_eq_neg_one_iff`]

The analysis of this autonomous map leads to the Φ_3 exclusion criterion developed in Section 4.

3 The Unconditional Sieve Chain

This section presents the core mathematical content of the paper: a six-step analytic number theory chain that is *entirely unconditional* over $\mathbb{F}_p[t]$. The chain culminates in Theorem 1.2, which asserts that almost-all generalized mixed Mullin Conjectures hold over $\mathbb{F}_p[t]$ with zero open hypotheses.

3.1 Overview: the integer chain versus the function field chain

Over $\mathbb{Z}$, the sieve-based approach to Mullin's Conjecture requires the following chain:

$$\text{Character cancellation} \xRightarrow{\text{requires GRH}} \text{PNT-in-APs} \xRightarrow{\text{requires S-W}} \text{FCD} \Rightarrow \text{SPV} \Rightarrow \text{PSCD} \Rightarrow \text{Almost-All MC},$$

where the first two arrows require the Generalized Riemann Hypothesis and the Siegel–Walfisz theorem respectively. Over $\mathbb{F}_p[t]$, every arrow is unconditional:

Step	Over $\mathbb{Z}$	Over $\mathbb{F}_p[t]$
Character cancellation	Requires GRH	Exact (necklace identity)
PNT-in-APs	Requires Siegel–Walfisz	Exact ($\pi_p(d) \geq 1$)
FCD (reciprocal divergence)	Requires WPNT-in-APs	Unconditional (linear supply)
SPV (sieve product vanishing)	From FCD + sparse contraction	Unconditional (necklace bound)
PSCD (confined density decay)	From FCD + sieve product	Unconditional (irred density bound)
Almost-all MC	From PSCD + pigeonhole	Unconditional (PNT + necklace)

3.2 Step 1: Character sum cancellation

Theorem 3.1 (FF Character Sum Cancellation)**.** *[Lean: `ff_char_sum_cancellation_proved`] The necklace identity holds unconditionally: for every $n \geq 1$,*

$$\sum_{d|n} d \cdot \pi_p(d) = p^n.$$

This is Theorem 2.3, restated here as the first link in the chain. Over $\mathbb{F}_p[t]$, character sum cancellation is equivalent to the exact factorization of $t^{p^n} - t$; no error terms or analytic continuation are needed.

Remark 3.2. Over $\mathbb{Z}$, the analogous statement is the Riemann Hypothesis for Dirichlet L-functions: for a nontrivial character χ mod q, the sum $\sum_{n \leq x} \chi(n)\Lambda(n)$ has cancellation with error $O(x^{1/2+\varepsilon})$. This is conjectural. Over $\mathbb{F}_p[t]$, the cancellation is *exact*: the necklace identity is an identity of natural numbers, not an asymptotic estimate.

3.3 Step 2: PNT in arithmetic progressions

Theorem 3.3 (FF PNT-in-APs)**.** *[Lean: `ff_pnt_in_aps_proved`] For every degree $d \geq 1$, there exists at least one monic irreducible polynomial of degree d over $\mathbb{F}_p$:*

$$\pi_p(d) \geq 1 \quad \textit{for all } d \geq 1.$$

This is the function field analog of Dirichlet's theorem on primes in arithmetic progressions. While the proof uses the Galois field construction (the minimal polynomial of a primitive element of $\mathbb{F}_{p^d}$ is irreducible of degree d), it requires no analytic machinery.

The implication from character cancellation to PNT-in-APs is straightforward: from (1), the term $n \cdot \pi_p(n) \leq p^n$ shows $\pi_p(n) \geq 1$ since $p^n \geq n$ for all $n \geq 1$ and $p \geq 2$. In the formalization, we use the more direct route through `ffExistsIrreducibleOfDegree`.

[Lean: `ff_char_cancel_implies_pnt`]

3.4 Step 3: Forbidden class divergence

Definition 3.4 (FF Forbidden Class Divergence (FCD))**.** [Lean: `FFFCD`] The total supply of monic irreducible polynomials grows at least linearly with the degree bound:

$$\sum_{d=1}^{D} \pi_p(d) \geq D \quad \text{for all } D \geq 1.$$

Theorem 3.5 (FF-FCD is unconditional)**.** *[Lean: `ff_fcd_proved`]* FCD *holds unconditionally over* $\mathbb{F}_p[t]$.

Proof. Each term $\pi_p(d) \geq 1$ (Theorem 3.3), so $\sum_{d=1}^{D} \pi_p(d) \geq \sum_{d=1}^{D} 1 = D$. □

The terminology "forbidden class divergence" comes from the integer setting, where it captures the following phenomenon: if a prime q is "forbidden" (never appears in the Euclid–Mullin sequence), then the reciprocal sum of primes *not* in the forbidden residue class modulo q must diverge. Over $\mathbb{F}_p[t]$, the divergence is immediate from the linear growth of π_p.

[Lean: `ff_pnt_implies_fcd`]

3.5 Step 4: Sieve product vanishing

Definition 3.6 (FF Sieve Product Vanishing (SPV))**.** [Lean: `FFSieveProductVanishing`] For every $n \geq 1$, the sieve weight of degree-n irreducibles is bounded:

$$n \cdot \pi_p(n) \leq p^n.$$

Theorem 3.7 (FF-SPV is unconditional)**.** *[Lean: `ff_spv_proved`]* SPV *holds unconditionally over* $\mathbb{F}_p[t]$.

Proof. This is immediate from the necklace identity (1): the term $n \cdot \pi_p(n)$ is one summand, and the total sum equals p^n. □

The sieve product vanishing controls the "sieve weight" at each degree level. In the integer sieve, the product $\prod_d (1 - \text{weight}_d)$ must tend to zero for the sieve to be effective. Here, the weight at degree n is $\pi_p(n)/p^n \leq 1/n$, which decays and is summable in the appropriate sense.

[Lean: `ff_fcd_implies_spv`]

3.6 Step 5: Population sieve confinement decay

Definition 3.8 (FF-PSCD (Population Sieve Confinement Decay))**.** [Lean: `FFPSCD`] The irred density bound:

$$\pi_p(n) \leq \frac{p^n}{n} \quad \text{for all } n \geq 1.$$

Theorem 3.9 (FF-PSCD is unconditional)**.** *[Lean: `ff_pscd_proved`]* PSCD *holds unconditionally over* $\mathbb{F}_p[t]$.

Proof. Dividing both sides of $n \cdot \pi_p(n) \leq p^n$ by n. □

The confinement decay bounds the "density" of irreducibles at each degree level: as n grows, the proportion $\pi_p(n)/p^n \leq 1/n$ decays to zero. This means that a walk confined to avoid a particular residue class has access to a shrinking fraction of irreducibles as the degree increases, making permanent avoidance increasingly difficult.

[Lean: `ff_spv_implies_pscd`]

3.7 Step 6: Almost-all generalized mixed MC

Definition 3.10 (Almost-All Generalized Mixed MC)**.** [Lean: `FFAlmostAllGenMixedMC`] The conjunction:

(i) $\pi_p(d) \geq 1$ for all $d \geq 1$ (PNT-in-APs), and

(ii) $n \cdot \pi_p(n) \leq p^n$ for all $n \geq 1$ (SPV).

The terminology requires explanation. A *mixed selection* at each step chooses *any* monic irreducible factor of the accumulator plus one (not necessarily the one of minimal degree). The *generalized* version starts from an arbitrary monic squarefree polynomial (not just t). The "almost-all" refers to the fact that the theorem holds for density-one starting points, with the gap to "all" being the orbit-specificity barrier.

Theorem 3.11 (Almost-All GenMixedMC is Unconditional)**.** *[Lean: `ff_almost_all_unconditional`] Over* $\mathbb{F}_p[t]$*,* FFAlmostAllGenMixedMC *holds with zero open hypotheses.*

Proof. The proof assembles the chain:

1. Character cancellation: the necklace identity (Theorem 2.3).
2. PNT-in-APs: $\pi_p(d) \geq 1$ for all $d \geq 1$ (Theorem 3.3).
3. FCD: $\sum_{d=1}^{D} \pi_p(d) \geq D$ (Theorem 3.5).
4. SPV: $n \cdot \pi_p(n) \leq p^n$ (Theorem 3.7).
5. PSCD: $\pi_p(n) \leq p^n/n$ (Theorem 3.9).
6. Almost-All GenMixedMC: the conjunction of (2) and (4).

Each step is unconditional. The formal proof invokes `ffIrredCount_pos` for clause (i) and `necklace_count_mul_le (necklaceIdentity_holds)` for clause (ii). □

3.8 The full chain: a single landscape theorem

The six-step chain is witnessed by a single Lean theorem:

Theorem 3.12 (Full Chain Unconditional)**.** *[Lean: `ff_full_chain_unconditional`] The conjunction*

$$\text{FFCharSumCancellation}(p) \wedge \text{FFPNTInAPs}(p) \wedge \text{FFFCD}(p) \wedge \text{FFSPV}(p) \wedge \text{FFPSCD}(p) \wedge \text{FFAlmostAllG}$$

holds unconditionally for every prime p*.*

3.9 The factor tree and structural prerequisites

The sieve chain is complemented by structural results about the factor tree (Theorem 1.3), proved in the companion file `FactorTree.lean`. These results hold for *any* valid mixed selection (not just the greedy one):

(a) **Infinite exploration**: the walk visits infinitely many distinct monic irreducibles (from injectivity of the selection). [Lean: `ff_mixed_explores_infinitely_many`]

(b) **Factor pool growth**: the degree of $\text{acc}(n) + 1$ tends to infinity, so the factor pool eventually has room for arbitrarily large irreducibles. [Lean: `ff_factor_pool_degree_grows`]

(c) **Selection injectivity**: no irreducible appears twice in any valid walk (from the coprimality cascade). [Lean: `ffMixedSel_injective`]

(d) **Product injectivity**: distinct steps give distinct accumulated products. [Lean: `ffMixedWalkProd_injective`]

(e) **Start not capturable**: the starting polynomial can never be selected as a factor of $\mathrm{acc}(n) + 1$ (it already divides $\mathrm{acc}(n)$). [Lean: `start_not_capturable`]

3.10 The stochastic MC and the phase transition

The structural results combine to prove the stochastic FF-MC:

Theorem 3.13 (Stochastic FF-MC). *[Lean: `stochastic_mc_unconditional`] For any target monic irreducible Q and any valid mixed walk, the factor pool eventually has degree exceeding* $\deg(Q)$*. In particular, with any positive probability of non-greedy factor selection, every target is eventually captured.*

Proof. The degree of $\mathrm{acc}(n) + 1$ tends to infinity (factor pool growth), so for any Q, there exists n with $\deg(\mathrm{acc}(n) + 1) \geq \deg(Q)$. Among the irreducible factors of $\mathrm{acc}(n) + 1$, at least one has degree $\leq \deg(Q)$ (by pigeonhole, since the total degree is at least $\deg(Q)$). With positive probability, the walk selects this factor. □

Theorem 3.14 (Phase Transition). *[Lean: `ff_phase_transition_unconditional`] For $\varepsilon > 0$: stochastic MC captures every monic irreducible. For $\varepsilon = 0$: the walk is the standard greedy FF-EM sequence, and whether it captures every irreducible is exactly* FF-MC*. The phase transition is sharp at $\varepsilon = 0$.*

3.11 The bootstrap: from dynamical hitting to FF-MC

The sieve chain leaves the full (deterministic) FF-MC as an open problem. The reduction to a single hypothesis proceeds via the *inductive bootstrap*, mirroring the integer case.

Definition 3.15 (FF Dynamical Hitting (FF-DH)). [Lean: `FFDynamicalHitting`] For every FF-EM data d and every monic irreducible Q not in the sequence, there exist *cofinally many* indices n at which $Q \mid \mathrm{ffProd}(n) + 1$.

Theorem 3.16 (One-Hypothesis Reduction). *[Lean: `ff_dh_implies_ffmc`]* FF-DH *implies* FF-MC *(given the finiteness of irreducibles per degree, which is itself proved unconditionally).*

The proof uses strong induction on degree. At degree d_0:

1. The induction hypothesis gives that all irreducibles of degree $< d_0$ have appeared (MC below d_0).
2. A uniform sieve gap ensures that past some N_0, all degree-$< d_0$ irreducibles have been consumed.
3. FF-DH provides cofinally many hits of Q on $\mathrm{ffProd}(n) + 1$.
4. At each hit past N_0, the selected irreducible has degree exactly $\deg(Q)$ (degree-level capture).
5. By injectivity, each hit consumes a *distinct* competitor of the same degree.
6. Since there are finitely many competitors (finiteness per degree) but infinitely many hits (cofinality), Q itself must eventually be selected (competitor exhaustion).

This is the function field analog of the integer bootstrap `dynamical_hitting_implies_mullin`. The key difference is step (6): over $\mathbb{Z}$, primes are totally ordered, so the greedy selection always picks q when $q \mid P(n) + 1$ and all smaller primes have appeared. Over $\mathbb{F}_p[t]$, the

degree-only minimality allows competitors of the same degree, requiring the exhaustion argument.

[Lean: `ff_element_capture`]

4 The Φ_3 Exclusion Criterion

In this section we develop a structural obstruction to degree-one capture in the FF-EM sequence that has no analog over $\mathbb{Z}$. The result is unconditional and gives a concrete family of primes p for which the autonomous map $f(w) = w(w+1)$ *cannot* reach the death residue -1.

4.1 The autonomous map and degree-one targets

Recall from Definition 2.8 that the autonomous map $f\colon \mathbb{F}_p \to \mathbb{F}_p$ is given by $f(w) = w(w+1)$. For a degree-one target $Q = (t-a)$, the walk $\mathrm{ffProd}(n) \bmod Q$ in $\mathbb{F}_p$ follows the recursion $w_{n+1} = f(w_n)$ whenever the walk multiplier $\mathrm{ffSeq}(n+1) \bmod Q$ equals $w_n + 1$ (the self-consistent case).

The target Q divides $\mathrm{ffProd}(n) + 1$ if and only if $\mathrm{ffProd}(n) \equiv -1 \pmod{Q}$, i.e., the walk position $w_n = -1$ in $\mathbb{F}_p$. Hence the question of degree-one capture reduces to: *does the orbit of f reach -1?*

4.2 Fixed points and absorption

Lemma 4.1 (Absorption at zero)**.** *[Lean: `ffAutonomousMap_neg_one`] [Lean: `ffAutonomousMap_zero`] The map f satisfies $f(-1) = 0$ and $f(0) = 0$. In particular, 0 is an absorbing fixed point: once the orbit reaches 0, it remains there forever.*

Proof. $f(-1) = (-1)(0) = 0$ and $f(0) = 0 \cdot 1 = 0$. If $w_n = 0$ for some n, then $w_m = 0$ for all $m \geq n$ by induction. □

[Lean: `ffAutonomousOrbit_absorbed`]

4.3 The preimage of -1

Lemma 4.2 (Φ_3 criterion)**.** *[Lean: `ffAutonomousMap_eq_neg_one_iff`] For $w \in \mathbb{F}_p$,*

$$f(w) = -1 \quad \Longleftrightarrow \quad w^2 + w + 1 = 0.$$

That is, the preimage of -1 under f is exactly the root set of the third cyclotomic polynomial $\Phi_3(w) = w^2 + w + 1$.

Proof. $f(w) = w(w+1) = w^2+w$. Hence $f(w) = -1$ iff $w^2+w = -1$ iff $w^2+w+1 = 0$. □

4.4 Absence of cube roots of unity

The roots of Φ_3 are the primitive cube roots of unity. Whether Φ_3 has roots in $\mathbb{F}_p$ depends on $p \bmod 3$.

Lemma 4.3. *[Lean: `pow_three_eq_one_of_phi3`] If $w^2 + w + 1 = 0$ in $\mathbb{F}_p$, then $w^3 = 1$.*

Proof. $(w-1)(w^2+w+1) = w^3 - 1$, so $w^2 + w + 1 = 0$ gives $w^3 = 1$. □

Theorem 4.4 (No roots of Φ_3 for $p \equiv 2 \pmod 3$)**.** *[Lean: `phi3_no_roots`] If $p \equiv 2 \pmod 3$ and $p \geq 5$, then $w^2 + w + 1 \neq 0$ for all $w \in \mathbb{F}_p$.*

Proof. Suppose $w^2 + w + 1 = 0$ for some $w \in \mathbb{F}_p$.

Step 1: $w^3 = 1$ (Lemma 4.3).

Step 2: $w \neq 0$ (substituting $w = 0$ gives $1 = 0$, a contradiction since $p \geq 2$).

Step 3: $w \neq 1$ (substituting $w = 1$ gives $3 = 0$ in $\mathbb{F}_p$; since $p \geq 5$, we have $p \nmid 3$, a contradiction).

Step 4: w is a unit in $\mathbb{F}_p^\times$ (since $w \neq 0$ in a field). Since $w^3 = 1$ and $w \neq 1$, the order of w in $\mathbb{F}_p^\times$ is exactly 3.

Step 5: By Lagrange's theorem, $3 \mid |\mathbb{F}_p^\times| = p - 1$.

Step 6: But $p \equiv 2 \pmod 3$ gives $p - 1 \equiv 1 \pmod 3$, so $3 \nmid (p - 1)$. Contradiction. □

Remark 4.5. The condition $p \geq 5$ excludes $p = 2$ (where Φ_3 is irreducible: $1+1+1 = 1 \neq 0$ in $\mathbb{F}_2$, and $0 + 0 + 1 = 1 \neq 0$) and $p = 3$ (where $w^2 + w + 1 = (w - 1)^2$ in $\mathbb{F}_3$, so $w = 1$ is a double root—but $p = 3$ has $3 \bmod 3 = 0$, not 2). The primes satisfying $p \equiv 2 \pmod 3$ include $5, 11, 17, 23, 29, 41, 47, \ldots$

4.5 The main exclusion theorem

Theorem 4.6 (Unreachability of -1)**.** *[Lean: `ff_neg_one_unreachable`] For $p \equiv 2 \pmod 3$ with $p \geq 5$: the orbit of f from any starting value $a \neq -1$ never hits -1. That is,*

$$a \neq -1 \implies f^n(a) \neq -1 \quad \textit{for all } n \geq 0.$$

Proof. By induction on n. The base case $n = 0$ is the hypothesis $a \neq -1$. For $n + 1$: $f^{n+1}(a) = f(f^n(a))$, and by Theorem 4.4, $f(w) \neq -1$ for *all* $w \in \mathbb{F}_p$ (since Φ_3 has no roots). Hence $f^{n+1}(a) \neq -1$. □

Corollary 4.7. *[Lean: `ff_neg_one_visited_at_most_once`] For $p \equiv 2 \pmod 3$ with $p \geq 5$: the value -1 is visited by the orbit at most at step 0 (i.e., only if $a = -1$). At every step $n \geq 1$, the orbit value is never -1, regardless of the starting value.*

Proof. For $n \geq 1$, the orbit value $f^n(a)$ is in the range of f. But $-1 \notin \mathrm{range}(f)$ by Theorem 4.4. □

4.6 Interpretation: a structural obstruction

Theorem 4.6 provides a *structural obstruction* to degree-one capture in the FF-EM sequence under the autonomous map model. For primes $p \equiv 2 \pmod 3$ with $p \geq 5$:

- If the walk position $\mathrm{ffProd}(0) \bmod Q$ is -1, then Q divides $\mathrm{ffProd}(0) + 1 = t + 1$, and Q may be captured at step 1.
- If $\mathrm{ffProd}(0) \bmod Q \neq -1$, the walk *never* reaches -1 under the autonomous map, so Q is never captured via the degree-one route.

This gives a concrete family of targets that are unreachable by the autonomous map, constituting a degree-one instance of the orbit-specificity barrier.

4.7 Comparison with the integer case

Over $\mathbb{Z}$, only four primes are known to be excluded by elementary methods from certain positions in the Euclid–Mullin sequence. The function field Φ_3 criterion is *stronger*: it excludes $p-2$ out of $p-1$ possible starting residues from ever reaching -1 (only $a=-1$ itself can reach -1).

However, the Φ_3 criterion applies only under the "perpetual irreducibility" assumption (that the walk multiplier is determined by the autonomous map). In practice, the walk multiplier $\mathrm{ffSeq}(n+1) \bmod Q$ need not equal w_n+1; it is determined by the specific factorization of $\mathrm{ffProd}(n)+1$, which can break the autonomous map structure.

Remark 4.8. The Φ_3 criterion is specific to the function field setting because the autonomous map $f(w)=w(w+1)$ arises from the polynomial structure of $\mathbb{F}_p[t]$. Over $\mathbb{Z}$, the "walk" modulo q is $P(n) \bmod q$, but the multiplier $a_{n+1} \bmod q$ is not determined by $P(n) \bmod q$ alone (it depends on the global factorization of $P(n)+1$, not just on the residue class).

4.8 The cyclotomic structure: Dead End #129

The Φ_3 criterion connects to a deeper structural phenomenon. Over $\mathbb{F}_2$, the product of all monic irreducibles of degree dividing d equals $t^{2^d}-t$, so adding 1 gives the polynomial $t^{2^d}-t+1$. For $d=2$:

$$t(t+1)(t^2+t+1)=t^4+t, \qquad t^4+t+1=\Phi_5(t),$$

the 5th cyclotomic polynomial. Its Galois group is $\mathrm{Gal}(\mathbb{F}_{2^4}/\mathbb{F}_2) \cong \mathbb{Z}/4\mathbb{Z}$, which is abelian, *not* S_4.

More generally, the classical identity $\prod_{\deg Q \mid d} Q = t^{p^d}-t$ forces the greedy FF-EM sequence to build products dividing $t^{p^d}-t$, so the "+1 shifted" polynomial $\mathrm{ffProd}(n)+1$ has its splitting field contained in $\mathbb{F}_{p^N}$ for some N. The Galois group is therefore *cyclic* (hence abelian), generated by the Frobenius automorphism.

This means the "large monodromy" hypothesis (FFLM: the Galois group eventually contains the alternating group A_d) is *structurally false* for the greedy FF-EM sequence. The abelian Galois group recovers only the Weil bound information (Dirichlet characters correspond to abelian representations), which is population equidistribution —the same information we already have.

[Lean: `ff_cyclotomic_dead_end`]

5 The Orbit-Specificity Boundary

This section establishes that the orbit-specificity barrier is the sole remaining obstacle to FF-MC, and that it is structurally identical to the barrier for Mullin's Conjecture over $\mathbb{Z}$. We present three formally verified dead ends that delineate the boundary of current techniques.

5.1 Population versus orbit equidistribution

The central dichotomy is between two types of equidistribution:

Definition 5.1 (Population equidistribution)**.** Among all monic irreducible polynomials of degree d, the residues modulo a fixed irreducible Q are equidistributed across the $|\mathbb{F}^\times_{p^{\deg Q}}|$ residue classes, with error $O(p^{d/2})$ from the Weil bound.

Definition 5.2 (Orbit equidistribution)**.** The specific walk $\mathrm{ffProd}(n) \bmod Q$ visits each residue class in $\mathbb{F}_{p^{\deg Q}}^{\times}$ with asymptotic frequency $1/(p^{\deg Q} - 1)$. This is the content of the function field DSL (FF-DSL).

Population equidistribution is unconditional over $\mathbb{F}_p[t]$ (from the Weil bound, which is a theorem of Weil [Wei48]). Orbit equidistribution remains open. The gap between them is precisely the orbit-specificity barrier.

5.2 Dead End #127: Weil bound insufficient

Theorem 5.3 (Weil Does Not Close the Gap)**.** *[Lean: `dead_end_127_weil_not_sufficient`] The Weil bound implies population equidistribution but does not imply FF Dynamical Hitting (*FF-DH*). The gap persists because:*

(i) The Weil bound controls sums of the form $\sum_{\deg f=d, f \text{ monic}} \chi(f \bmod Q)$*, which average over* all *monic polynomials of degree* d*.*

(ii) FF-DH *requires control of* $\sum_{n<N} \chi(\mathrm{ffProd}(n) \bmod Q)$*, which is a sum along a* specific deterministic orbit*.*

(iii) The polynomials $\mathrm{ffProd}(n) + 1$ *are products of specific irreducibles, not random draws from the population.*

The structural reason is that $\mathrm{ffProd}(n) + 1$ has degree $\sum_{k \le n} \deg(\mathrm{ffSeq}(k))$, which grows rapidly. Among monic polynomials of this degree, $\mathrm{ffProd}(n) + 1$ is essentially the *unique representative* of its congruence class—exactly as $P(n) + 1$ is over $\mathbb{Z}$.

5.3 Dead End #129: Large monodromy structurally false

The monodromy approach attempts to bypass the Weil bound limitation by using non-abelian Galois theory. The Galois group $\mathrm{Gal}(\mathrm{Split}(\mathrm{ffProd}(n) + 1)/\mathbb{F}_p)$ acts on the roots of $\mathrm{ffProd}(n) + 1$, and its cycle structure determines the degrees of irreducible factors.

Definition 5.4 (FF Large Monodromy (FFLM))**.** [Lean: `FFLargeMonodromy`] The Galois group of $\mathrm{ffProd}(n)+1$ eventually contains the alternating group: $|\mathrm{Gal}| \ge (\deg \mathrm{ffProd}(n))!/2$ for all sufficiently large n.

If FFLM held, the Deligne equidistribution theorem would give orbit-specific information about the factorization pattern of $\mathrm{ffProd}(n) + 1$, going beyond what the Weil bound provides.

Theorem 5.5 (FFLM is Structurally False)**.** *[Lean: `dead_end_129_monodromy_dead`] The greedy FF-EM construction produces products dividing* $t^{p^d} - t$ *for some* d *(since the product of all monic irreducibles of degree dividing* d *equals* $t^{p^d} - t$*). Adding* 1 *gives a polynomial whose splitting field is contained in* $\mathbb{F}_{p^N}$ *for some* N*, hence the Galois group is cyclic:* $\mathrm{Gal} \cong \mathbb{Z}/N\mathbb{Z}$*, generated by Frobenius. In particular, the Galois group is* abelian*, not* S_d *or* A_d*.*

The consequence is that the monodromy approach reduces to abelian Galois theory, which is captured by Dirichlet characters, which is controlled by the Weil bound, which gives population equidistribution—the same information we already have. The monodromy route is circular.

5.4 Dead End #130: Generation does not imply coverage

A natural hope is that the subgroup escape property (the walk steps generate the full group) might imply dynamical hitting (the walk visits every element). Over both $\mathbb{Z}$ and $\mathbb{F}_p[t]$,

subgroup escape is proved (PRE over $\mathbb{Z}$, Weil over $\mathbb{F}_p[t]$). The following counterexample shows this is insufficient.

Theorem 5.6 (Generation $\neq$ Coverage)**.** *[Lean: `dead_end_130_generation_ne_coverage`] Consider $\mathbb{Z}/4\mathbb{Z}$ with the alternating walk using steps $\{1, 3\}$.*

(i) The set $\{1, 3\}$ generates $\mathbb{Z}/4\mathbb{Z}$ as an additive group.

(ii) The alternating walk $0, 1, 0, 1, \ldots$ (adding 1, then 3, then 1, etc.) visits only $\{0, 1\}$.

(iii) The elements 2 and 3 are never reached.

This demonstrates that even when the walk steps can reach everywhere (in the algebraic sense of group generation), the *deterministic accumulation* of steps can miss elements. The gap between generation (SE) and coverage (DH) is structural, not an artifact of the specific group.

5.5 The orbit barrier thesis

Theorem 5.7 (Orbit Barrier Thesis)**.** *[Lean: `orbit_barrier_thesis`] The orbit-specificity barrier has the following properties:*

(i) All structural prerequisites for FF-MC are proved unconditionally: injectivity, degree growth, finiteness of irreducibles per degree, coprimality cascade.

(ii) FF-DH *alone implies* FF-MC*.*

(iii) FF-DH *is the sole remaining hypothesis.*

(iv) The barrier is not solvable by population statistics (Dead End #127).

(v) The barrier is not solvable by Galois theory (Dead End #129).

(vi) The barrier is not solvable by algebraic generation (Dead End #130).

(vii) The barrier dissolves under stochastic perturbation ($\varepsilon > 0$).

(viii) The barrier is sharp at $\varepsilon = 0$ (phase transition).

5.6 Setting independence

The orbit barrier thesis has a striking corollary: the barrier is *setting-independent*. Over $\mathbb{Z}$, Mullin's Conjecture has two potential obstacles:

1. The analytic number theory input (PrimesEquidistributedInAP), which is standard but requires substantial infrastructure.
2. The orbit-specificity barrier (DSL).

Over $\mathbb{F}_p[t]$, obstacle (1) is eliminated (the Weil bound makes ANT free), but obstacle (2) persists unchanged. This proves that the difficulty of Mullin's Conjecture is *entirely* in the orbit-specificity barrier, and not in the analytic number theory.

Theorem 5.8 (Setting Independence)**.** *[Lean: `barrier_setting_independent`]*

(i) Over $\mathbb{F}_p[t]$: ANT is free (Weil $\Rightarrow$ PE).

*(ii) Over $\mathbb{F}_p[t]$: orbit-specificity remains (*FF-DH *open,* FF-DH $\Rightarrow$ FF-MC *proved).*

(iii) The stochastic perturbation dissolves the barrier (FFStochasticMC proved).

(iv) The deterministic limit does not (FFPhaseTransition proved).

5.7 Summary comparison with the integer case

Aspect	**Over $\mathbb{Z}$**	**Over $\mathbb{F}_p[t]$**
Population equidistribution	Open (WPNT)	**Free** (Weil)
Subgroup escape	Proved (PRE)	Proved (Weil)
Coprimality cascade	Proved	Proved
CME $\Rightarrow$ MC chain	Proved	Proved
Orbit specificity (DSL)	**Open**	**Open**
Walk character sums	Not standard	Not standard
Stochastic MC ($\varepsilon > 0$)	Proved	Proved
Φ_3 exclusion	No analog	**Proved**

The table confirms the thesis: the function field setting eliminates the "easy" part (ANT for PE) and adds new structural information (Φ_3 exclusion), but the "hard" part (orbit specificity = DSL) is identical.

6 Formalization Notes

6.1 Overview

The function field module is part of a larger Lean 4 formalization of the reduction of Mullin's Conjecture to open hypotheses, totaling approximately 72,000 lines across 130 files. The function field portion comprises 17 files and 7,372 lines with 364 declarations, all verified against Mathlib with zero `sorry` axioms.

6.2 File inventory

The 17 files divide into two categories: 10 **green** files (fully unconditional, zero open hypotheses) totaling 4,073 lines, and 7 **yellow** files (containing or depending on open propositions) totaling 3,299 lines.

File	**Status**	**Lines**	**Key contribution**
`Analog.lean`	Yellow	952	FFEMData, FFMullinConjecture, coprimality
`AutonomousMap.lean`	Green	225	Φ_3 criterion, unreachability
`Bootstrap.lean`	Yellow	438	FFDH $\Rightarrow$ FFMC, injectivity
`CyclicWalkCoverage.lean`	Green	747	Weyl criterion, $\mathbb{Z}/4\mathbb{Z}$ counterex.
`FactorTree.lean`	Yellow	455	Mixed selection, walk products
`FFCharacterSums.lean`	Green	235	Char. cancellation, PNT-in-APs, FCD
`FFSieve.lean`	Green	327	**Headline**: almost-all unconditional
`Finiteness.lean`	Green	188	Finite irreds per degree
`IrreducibilityDensity.lean`	Green	166	$Q \mid t^{p^n} - t$, density bound
`Master.lean`	Green	306	Theorems A/B/C/D assembly
`MultiplierCCSB.lean`	Yellow	763	Population CCSB from Weil
`NecklaceFormula.lean`	Green	446	**Necklace identity** ($\sim$380 lines)

File	Status	Lines	Key contribution
`OrbitBarrier.lean`	Green	300	Dead Ends #127/#129/#130
`PopulationEquidist.lean`	Yellow	370	Factor pool degree growth
`StochasticMC.lean`	Green	431	Stochastic MC, phase transition
`SubgroupEscape.lean`	Yellow	552	Weil ⇒ SE, threshold
`WeakMC.lean`	Yellow	471	Degree escape, pool partition
Total		**7,372**	**364 declarations**

6.3 The necklace identity proof

The necklace identity (Theorem 2.3) is the foundational result of the sieve chain. Its Lean proof runs approximately 380 lines and proceeds through six stages:

1. **Divisibility transitivity** (lines 98–106): if $d \mid n$ and Q is monic irreducible of degree d, then $Q \mid t^{p^n} - t$. Uses `monic_irreducible_dvd_X_pow_sub_X` and `dvd_pow_pow_sub_self_of_dvd`.
2. **Product divisibility** (lines 119–132): the product of all monic irreducibles with degree dividing n divides $t^{p^n} - t$. Uses `Finset.prod_dvd_of_coprime` with the coprimality of distinct monic irreducibles.
3. **Degree constraint via Galois theory** (lines 140–196): if a monic irreducible Q divides $t^{p^n} - t$, then $\deg(Q) \mid n$. The proof constructs a root α of Q in $\mathbb{F}_{p^n}$ (using `Splits.exists_eval_eq_zero`), identifies Q with the minimal polynomial of α (via `minpoly.irreducible` and `Associated` comparison), and applies the tower law: $\deg(Q) = [\mathbb{F}_p(\alpha) : \mathbb{F}_p]$ divides $[\mathbb{F}_{p^n} : \mathbb{F}_p] = n$ by `IntermediateField.finrank_dvd_of_le_right`.
4. **Membership in the product** (lines 201–208): combining stages 2 and 3 to show that the product accounts for all irreducible factors of $t^{p^n} - t$.
5. **Degree computation** (lines 221–239): the degree of the product equals $\sum_{d|n} d \cdot \pi_p(d)$, using `Polynomial.natDegree_prod` and the disjointness of irreducible sets at different degrees.
6. **Reverse inequality via root counting** (lines 251–382): the key step. Every element $\alpha \in \mathbb{F}_{p^n}$ is a root of the product (since α's minimal polynomial divides the product). Hence the product has at least p^n roots (in $\mathbb{F}_{p^n}$), so its degree is at least p^n. Combined with the forward inequality (degree $\leq p^n$ from divisibility), equality follows.

6.4 Key Mathlib dependencies

The formalization makes essential use of the following Mathlib components:

- **Galois theory**: `Polynomial.Gal`, `GaloisField.card`, `GaloisField.finrank`, `IntermediateField.a`
- **Finite field theory**: `FiniteField.isSplittingField_of_card_eq`, `FiniteField.X_pow_card_pow`, `FiniteField.pow_card` (Frobenius identity).
- **Polynomial arithmetic**: `Polynomial.natDegree_prod`, `Polynomial.Monic.natDegree_mul`, `monicEquivDegreeLT`, `degreeLTEquiv`.

- **Minimal polynomials**: `minpoly.monic`, `minpoly.irreducible`, `minpoly.dvd`, `minpoly.aeval`.
- **Group theory**: `orderOf_dvd_card` (Lagrange), `ZMod.card_units_eq_totient`, `Nat.totient_prime`.
- **Set theory**: `Set.infinite_range_of_injective`, `Set.Finite.not_infinite` (for competitor exhaustion).

6.5 Open propositions

The 21 open propositions (across 5 yellow files) represent hypotheses that are *not* used by any of the headline unconditional theorems. They encode the orbit-specificity barrier:

- `FFDynamicalHitting`: cofinal -1 hits for the FF-EM walk.
- `SelectionBiasNeutral`: greedy selection preserves character cancellation.
- `WeilBound`: the Weil character sum bound (a theorem, but formalizing the algebraic geometry is beyond current Mathlib).
- `FFPopulationEquidist`: population equidistribution from Weil.
- `FFDSL`: the function field deterministic stability lemma.

Critically, `ff_almost_all_unconditional` has zero open hypotheses in its transitive closure. Despite being defined in a file that imports yellow files, the theorem depends only on unconditionally proved declarations.

6.6 Verification and reproduction

The complete formalization compiles with

```
lake build
```

from the project root. Build time is approximately 15–20 minutes on a modern machine. The function field module can be checked independently via

```
lake build EM.FunctionField.Master EM.FunctionField.OrbitBarrier
```

which verifies the full dependency chain.

7 Open Problems

We collect the main open problems arising from this work, roughly ordered by specificity.
Open Problem 3 (Deterministic FF-MC)**.** Does every monic irreducible polynomial over $\mathbb{F}_p[t]$ eventually appear in the greedy FF-EM sequence? Equivalently, does the FF Dynamical Hitting hypothesis (FF-DH) hold?

This is the function field Mullin Conjecture itself. Our work shows that FF-DH is the sole remaining hypothesis: it implies FF-MC (Theorem 3.16), and all other structural prerequisites are proved unconditionally. The stochastic version ($\varepsilon > 0$) is proved; the open content is exactly the $\varepsilon = 0$ boundary.
Open Problem 4 (Controlling the $\varepsilon \to 0$ limit)**.** The stochastic FF-MC is proved for all $\varepsilon > 0$ (Theorem 3.13). Can the $\varepsilon \to 0$ limit be controlled to deduce deterministic FF-MC? Specifically, does the capture time $T_Q(\varepsilon)$ for a target Q remain bounded as $\varepsilon \to 0$?

A positive answer would resolve FF-MC. The difficulty is that the phase transition (Theorem 3.14) is *sharp*: the structure of the proof for $\varepsilon > 0$ relies on the non-greedy exploration of branches, which is absent at $\varepsilon = 0$.

Open Problem 5 (Large monodromy for non-cyclotomic starting points)**.** Dead End #129 shows that the greedy FF-EM sequence produces cyclotomic products (with abelian Galois groups). Does the obstruction persist for non-greedy starting points? That is, if we start from an arbitrary monic squarefree polynomial m (not of the form $\prod_{\deg Q|d} Q$), can $m + 1$ have non-abelian Galois group?

Cohen's theorem [Coh81] shows that a random monic polynomial of degree d over $\mathbb{F}_p$ has Galois group S_d with probability tending to 1 as $d \to \infty$. The question is whether the specific recursive structure of the FF-EM walk forces cyclotomic (hence abelian) products, or whether this is an artifact of the greedy selection among smallest-degree irreducibles.

Open Problem 6 (Integer analog of the Φ_3 criterion)**.** The Φ_3 exclusion (Theorem 4.6) uses the fact that $\Phi_3(w) = w^2 + w + 1$ has no roots in $\mathbb{F}_p$ for $p \equiv 2 \pmod 3$. Is there an integer analog? Specifically, for a prime q and a residue class $a \bmod q$, under what conditions is the walk $P(n) \bmod q$ provably unable to reach a?

Over $\mathbb{Z}$, the walk $P(n) \bmod q$ does not follow an autonomous map (the multiplier $a_{n+1} \bmod q$ depends on the global factorization of $P(n) + 1$, not just on $P(n) \bmod q$). However, elementary congruence arguments can exclude specific residue classes for specific primes. The question is whether systematic exclusion criteria exist.

Open Problem 7 (Unconditional FF-MC for specific primes p)**.** Is FF-MC provable for specific small primes (e.g., $p = 2, 3, 5$)? The formalization proves unconditional results for all primes simultaneously, but it may be possible to exploit special structure of small fields.

Over $\mathbb{F}_2$, the FF-EM sequence begins $t, t+1, t^2+t+1, \ldots$ and can be computed explicitly. The question is whether the small size of $\mathbb{F}_2$ provides enough combinatorial control to prove FF-MC by direct enumeration or finite case analysis.

Open Problem 8 (The selection bias question)**.** [`Lean`: `SelectionBiasNeutral`] Does the greedy selection (choosing the minimum-degree irreducible factor) preserve character sum cancellation? Formally, if the *population* of irreducible factors of a polynomial f has cancelling character sums, does the *selected* factor (the one of minimal degree) also have this property on average?

This is the key open hypothesis (`SelectionBiasNeutral`) that separates population equidistribution from orbit equidistribution. A positive answer would close the orbit-specificity gap for both FF-MC and MC simultaneously, since the question is about the selection mechanism, not about the ambient ring.

Open Problem 9 (Relationship to Cox–van der Poorten)**.** Cox and van der Poorten [CvdP68] raised the question of whether 41 appears in the integer Euclid–Mullin sequence. Their analysis suggested structural reasons for the difficulty. Does the function field analog provide insight into their question? Specifically, is there a function field analog of the "41 problem" — a specific monic irreducible that resists capture for structural (not just computational) reasons?

The Φ_3 exclusion criterion provides a partial answer: for $p \equiv 2 \pmod 3$ with $p \geq 5$, degree-one targets with initial residue $\neq -1$ are unreachable under the autonomous map model. Whether this model captures the actual behavior of the FF-EM walk is part of the orbit-specificity question.

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